CH5440 : Assignment 3 : Principal Component Analysis

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QUESTION 1 :

$$\bar{x} = \begin{bmatrix} 9 \\ 68 \\ 129 \end{bmatrix} \qquad S = \begin{bmatrix} 7 & 21 & 34 \\ 21 & 64 & 102 \\ 34 & 102 & 186 \end{bmatrix}$$

Largest eigenvalue = 250.4009

trace of matrix = sum of eigenvalues

$$7+64+186 = \lambda_1+\lambda_2+250.4009$$

$$6.5991 = \lambda_1+\lambda_2$$

determinant of matrix = product of eigenvalues

$$\begin{vmatrix} 7 & 21 & 34 \\ 21 & 64 & 102 \\ 34 & 102 & 186 \end{vmatrix} = \lambda_1\lambda_2(250.4009) = 146$$

$$\lambda_1\lambda_2 = 0.583065$$

$$\Rightarrow 6.5991 = \lambda_1 + \frac{0.583065}{\lambda_1} \quad \Rightarrow \lambda_1 = \frac{6.5991 \pm \sqrt{6.5991^2 - 4(0.583065)}}{2}$$

$$\lambda_1 = 3.2995 \pm 3.2099$$

therefore

$$\lambda_1 = 6.50945$$

$$\lambda_2 = 0.08965$$

(a) Calculating eigenvectors corresponding to eigenvalues

$$\begin{bmatrix} 7 & 21 & 34 \\ 21 & 64 & 102 \\ 34 & 102 & 186 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \lambda\begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

- $(7-\lambda)a + 21b + 34c = 0$
- $21a + (64-\lambda)b + 102c = 0$
- $34a + 102b + (186-\lambda)c = 0$

$$\left(\frac{21}{7-\lambda} - \frac{(64-\lambda)}{21}\right)b + \left(\frac{34}{7-\lambda} - \frac{102}{21}\right)c = 0$$

$$\Rightarrow (441 - 7(64) - \lambda^2 + 71\lambda)b = c(102(7) - 102\lambda - 21\times 34)$$

$$\Rightarrow -(\lambda^2 - 71\lambda + 7)b = c(-102\lambda)$$

$$\Rightarrow c = \left(\frac{\lambda^2 - 71\lambda + 7}{102\lambda}\right)b$$

now

$$34a + 102b + \frac{(186-\lambda)(\lambda^2-71\lambda+7)}{102\lambda}b = 0$$

$$a = \frac{1}{34\times 102\lambda}\left(\lambda^3 - \lambda^2(257) + 2809\lambda - 1302\right)$$

for $\lambda = 250.4009$

$a = 0.332b$, $c = 1.9591b$

normalized eigenvector $\equiv [0.162 \quad 0.4877 \quad 0.8578]^T$

for $\lambda = 6.50945$

$a = 0.2821b$, $c = -0.622b$

normalized eigenvector $\equiv [0.2331 \quad 0.8258 \quad -0.5135]^T$

for $\lambda = 0.08965$

$a = -3.8844b$, $c = 0.0703b$

normalized eigenvector $\equiv [-0.9588 \quad 0.2833 \quad 0.01992]^T$

(b) as we know that the eigenvalues represent the amount of variance captured by its corr. eigenvector

testing by retaining $\lambda = 250.4009$ $\Rightarrow$ % variance captured $= \dfrac{250.4009}{250.4009 + 0.08965 + 6.5095} = 0.9743$

$\Rightarrow$ eigenvector corr. to 250.4009 captures 97.43% of total variance hence only this eigenvector needs to be retained

(c) $$\begin{bmatrix} 0.2331 & 0.8258 & -0.5135 \\ -0.9588 & 0.2833 & 0.01992 \end{bmatrix} \begin{bmatrix} m \\ SVL \\ HLS \end{bmatrix} = 0$$

⇒ a better linear relationship would be using eigenvector corr. to eigenvalue closest to zero

$$\Rightarrow m(-0.9588) + SVL(0.2833) + HVL(0.01992) = 0$$

(d) Score calculation

data matrix : $[10.1 \quad 73 \quad 135.5]^T$

$$\begin{bmatrix} 0.162 & 0.4877 & 0.8578 \end{bmatrix} \begin{bmatrix} 10.1 \\ 73 \\ 135.5 \end{bmatrix} = 153.4702$$

(e) mass = ? SVL = 73 mm

from the matrix equation in (c) ⇒ $$\begin{bmatrix} 0.2331 & 0.8258 \\ -0.9588 & 0.2833 \end{bmatrix} \begin{bmatrix} m \\ SVL \end{bmatrix} = \begin{bmatrix} 0.5135 \\ -0.01992 \end{bmatrix} [HVL]$$

therefor after solving the above equation

$m = 13.779$ g $\quad SVL = 415.033$ mm

(f) measured SVL = 73 mm and measured HLS = 135.5 mm , mass = ?

$$m(-0.9588) + 73(0.2833) + 135.5(0.01992) = 0$$

$\therefore$ mass (m) = 24.385 grams

QUESTION 2 :

- **Euclidean Distance** in my solution is calculated as the **sum of absolute differences between the data vector and the reference vector** [normal image vector]
- Now for identifying the image we take the **Euclidean Distance** between the **data matrix** and the **normal image vector** corresponding each person and find the images that have the **minimum Euclidean Distance** while also being the correctly identified image
- This process is repeated for all the **15 normal images** and the correctly identified images are summed up to find the total number of identified images

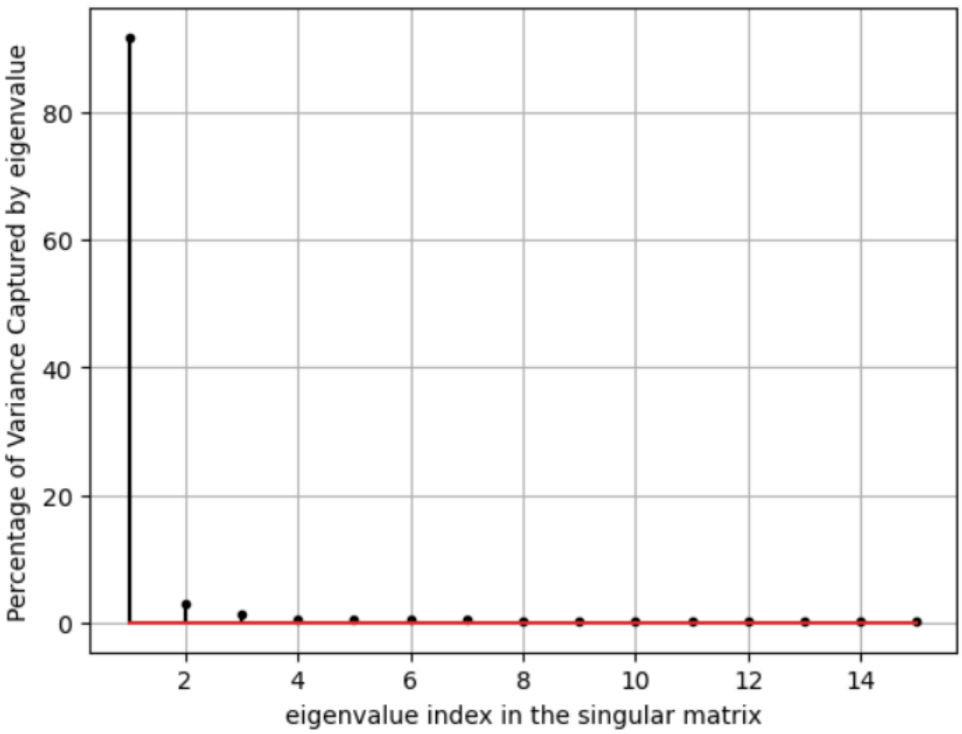
By performing this process I was able to identify **61 out of the 90** images in the data matrix correctly using their normal images

Now this data matrix is of dimensions **77760 x 90 storing such a huge volume of data** and then trying to identify them using **Euclidean Distance** or **any other algorithm will take a lot of time**.

So we **reduce the dimensionality of the data** by finding the **eigenvalues** and the **eigenvectors** of the **covariance matrix** of the data matrix. But we have a problem here, the **Covariance matrix** that we are interested in will have dimensions **77760 x 77760.**

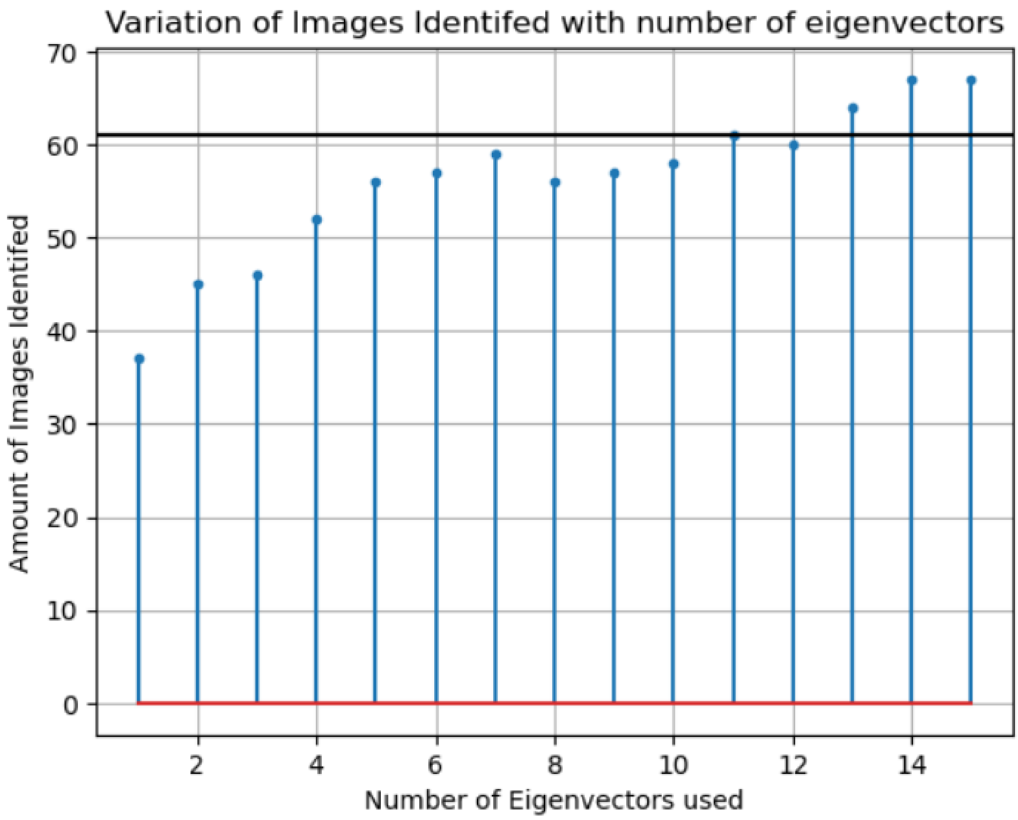
Computers will struggle to find their eigenvalues and eigenvectors because their huge size and will lead to memory limit exceeded error. Hence we use **Singular Value Decomposition (SVD)** on the data matrix to obtain the eigenvalues and the eigenvectors of the Covariance Matrix

The plot below shows the percentage variance capture by each Eigenvalue



Eigenvalue index	Percentage Variance Explained	Eigenvalue index	Percentage Variance Explained	Eigenvalue index	Percentage Variance Explained
1	91.706	6	0.4721	11	0.2096
2	3.041	7	0.4165	12	0.1722
3	1.398	8	0.3721	13	0.1636
4	0.6565	9	0.2979	14	0.1355
5	0.5644	10	0.2749	15	0.1195

- From the above Table we can see that the first **3 eigenvalues capture more than 95 % of the variance in the data**
- Now let us calculate the scores of the data matrix corresponding to varying number of eigenvectors and try to detect the images from the normal scores using the **Euclidean Distance Method**
- The number of images being identified by varying the number of eigenvectors are plotted below, Here the **black line represents the images identified by without data compression**



- We can observe that by using all the 15 eigenvectors we are capable of identifying **67 out of 90** images correctly, which is better than what we had by trying to identify the images without performing data compression
- Essentially we are operating on a **15 x 90** matrix instead of using a **77760 x 90** matrix to compute the Euclidean Distances

QUESTION 3 :

1. Three Linearly Independent Constraints exist between the 5 flow variables
2. The Eigenvalues and corresponding Eigenvectors are tabulated below :

```
EIGENVALUES AND EIGENVECTORS :
----------------------------------------------------------------------------------------------------
1 | 8.67843     | Eigenvector : [[ 0.12277361 -0.58112169  0.06381017  0.18830573 -0.76976835  0.12308585]]
2 | 3.61457     | Eigenvector : [[-0.41906663 -0.50530519 -0.17421795 -0.59463921  0.08736405 -0.42127148]]
3 | 0.96194     | Eigenvector : [[-0.43691843  0.14820627  0.70895487  0.26900453 -0.1277576  -0.44253219]]
4 | 0.01035     | Eigenvector : [[-0.64645944  0.41966047 -0.28325179 -0.13664477 -0.41880558  0.36286885]]
5 | 0.00974     | Eigenvector : [[-0.44346887 -0.45126681 -0.0168924   0.47207596  0.45062758  0.41651812]]
6 | 0.00996     | Eigenvector : [[ 0.06194869 -0.07240512  0.61841929 -0.54502681  0.0709851   0.55351994]]
----------------------------------------------------------------------------------------------------
```

The results after performing statistical Testing are tabulated below

$$Az_i^* = 0$$

- z_i^* : **True value of measurements of i th sample**
- A : **mxp constraint matrix**
- p : **Number of Variables**
- N : **Number of Samples in data Matrix**
- m : **Number of Contraints**

$$H_0 : \lambda_{N,p-d+1} = \lambda_{N,p-d+2} = \ldots = \lambda_{N,p}$$

$$H_1 : \textbf{at least two smallest d eigenvalues are unequal}$$

- The Statistic to test the Hypothesis is :

$$\tau = n' \cdot \left[d \cdot \ln \bar{l} - \sum_{j=p-d+1}^{j=p} \ln l_{N,j} \right]$$

$$\bar{l} = \frac{1}{d} \cdot \sum_{j=p-d+1}^{j=p} l_{N,j}$$

$$n' = N - \frac{2p+11}{6}$$

- Here $l_{N,j}$ is the jth eigenvalue of S_{z_s}
- The NULL HYPOTHESIS is rejected if the $\tau \geq \chi^2_{\nu,\alpha}$
- Here χ^2 is the chi-square distirbution with ν degrees of freedom, computed as $\nu = \frac{1}{2} \cdot (d+2) \cdot (d-1)$

```
STATISTICAL TESTING :
----------------------------------------------------
Chi-square at 95% and 14 dof            : 23.6848
The computed statistic value is         : 12109.1004
----------------------------------------------------
Chi-square at 95% and 9 dof             : 16.919
The computed statistic value is         : 8241.373
----------------------------------------------------
Chi-square at 95% and 5 dof             : 11.0705
The computed statistic value is         : 0.951
----------------------------------------------------

Number of constraints after performing the above test is : 3
```

We can see that the Null hypothesis is satisfied when the degrees of freedom becomes 5 or d = 3 [i.e 3 eigenvectors are needed to find the constraint matrix]

- The result obtained from hypothesis testing matches with that of the original constraint matrix's dimensions

```
EIGENVALUES AND EIGENVECTORS AFTER PERFORMING STATISTIC TESTING:
-------------------------------------------------------------------------------------------------------------
4 | 0.01035     | Eigenvector : [[-0.64645944  0.41966047 -0.28325179 -0.13664477 -0.41880558  0.36286885]]
5 | 0.00974     | Eigenvector : [[-0.44346887 -0.45126681 -0.0168924   0.47207596  0.45062758  0.41651812]]
6 | 0.00996     | Eigenvector : [[ 0.06194869 -0.07240512  0.61841929 -0.54502681  0.0709851   0.55351994]]
-------------------------------------------------------------------------------------------------------------
```

3. The Original Constraint matrix and the Estimated Constraint matrix are as follows :

```
[[-1  1 -1  0 -1  0]
 [ 0 -1  0  1  1  0]
 [ 0  0  1 -1  0  1]]
```

Original Matrix

```
[[-0.64645944  0.41966047 -0.28325179 -0.13664477 -0.41880558  0.36286885]
 [-0.44346887 -0.45126681 -0.0168924   0.47207596  0.45062758  0.41651812]
 [ 0.06194869 -0.07240512  0.61841929 -0.54502681  0.0709851   0.55351994]]
```

Estimated Matrix

The Subspace Angles between the Original and Estimated Matrices are as follows

```
-----------------------------
subspace angle 1 : 5.0037e-16
subspace angle 2 : 1.9008e-16
subspace angle 3 : 1.6906e-16
-----------------------------
```

4. Inorder to determine the poor set of independent variables we decompose the constraint matrix into Ad and Ai, where Ad contains the coefficients corresponding to the dependent variables and Ai contains the coefficients corresponding to the independent variables.

   Now inorder to determine all the dependent variables in our flow simulation Ad must be invertible, for that to happen Ad must be non singular or its condition number [maximum eigenvalue / minimum eigenvalue] should be less than 1000

   Hence we iterate through all the possibilities for set of dependent variables and then check their condition numbers, the ones that have very high condition number are the poor choices for dependent variables

   From the code we can observe that the combinations **F1,F2 and F5 or F3,F4 and F5 yield very high condition numbers** hence these variables cannot be dependent variables **[2 out of the 20 combinations are poor choices]**

5. True Regression matrix :

```
[[ 0.  0. -1.]
 [-1. -1.  0.]
 [-1.  0.  1.]]
```

Estimated Regression Matrix :

```
[[  0.21137408  -0.99796289  -1.28108229]
 [ -1.06450754  -0.99858347 -24.65712951]
 [ -8.79803597  -0.98038794   0.89505607]]
```

Absolute Difference between the True and Estimated Regression matrix :

```
[[2.11374084e-01 9.97962894e-01 2.81082290e-01]
 [6.45075439e-02 1.41653123e-03 2.46571295e+01]
 [7.79803597e+00 9.80387939e-01 1.04943933e-01]]
```